AUTHOR TESTED VIDEO PROOF PROVIDED

ADC And DAC TRAINER

A. ASOKAN AMBALI AND CHINCHU GRACE LUKOSE

Analogue to digital converter (ADC) and digital to analogue converter (DAC) are integral parts of data processing and widely used industrial and household electronic devices. Different conversion methods include resistor ladder, binary weightage, and cascaded op-amp.

Most trainer kits available in the market start showing wrong results after a few years of usage due to the aging of components. This device would work accurately for many more years.

The project uses Arduino IDE as the software development platform, so the trainees can perform ADC and DAC experiments as well as get an idea about the configurations of ADC and DAC in Arduino. The main advantages of Arduino software are easy editing and debugging.

Circuit and working

Fig. 1 shows circuit diagram of the DAC-ADC trainer. The circuit consists of ATmega 328P microcontroller interfaced with 20x4 LCD module and I2C module, 8-bit LED digital word display/generator, voltage regulator, voltage divider circuit for ADC input, and nine switching diodes, besides some discrete components such as resistors and capacitors.

The ATmega 328P microcontroller used in Arduino Uno has six inbuilt ADCs with 10-bit resolution and six DACs with 8-bit resolution that use pulse width modulation (PWM) for DAC conversion. The Arduino IDE is known to be the most popular and cost-effective software development platform.

This circuit using ATmega 328P microcontroller can be seen at *www.arduino.cc*. The ADC input can be from any of the six inbuilt ADCs in the chip.

In the circuit, 9V-0-9V AC output of 230V AC step-down transformer X1 is rectified by diodes D11 and D12 and smoothened by capacitors C1 and C2. This 9V DC is connected to the input of regulator IC 7805, which converts it into 5.6V through diode D9. Diode D10 converts it to 5V, which is used as positive supply to the circuit. Filter capacitors C3, C4, and C5 smoothen the supply voltage.

During operation, the power supply may fall slightly below 5V. The

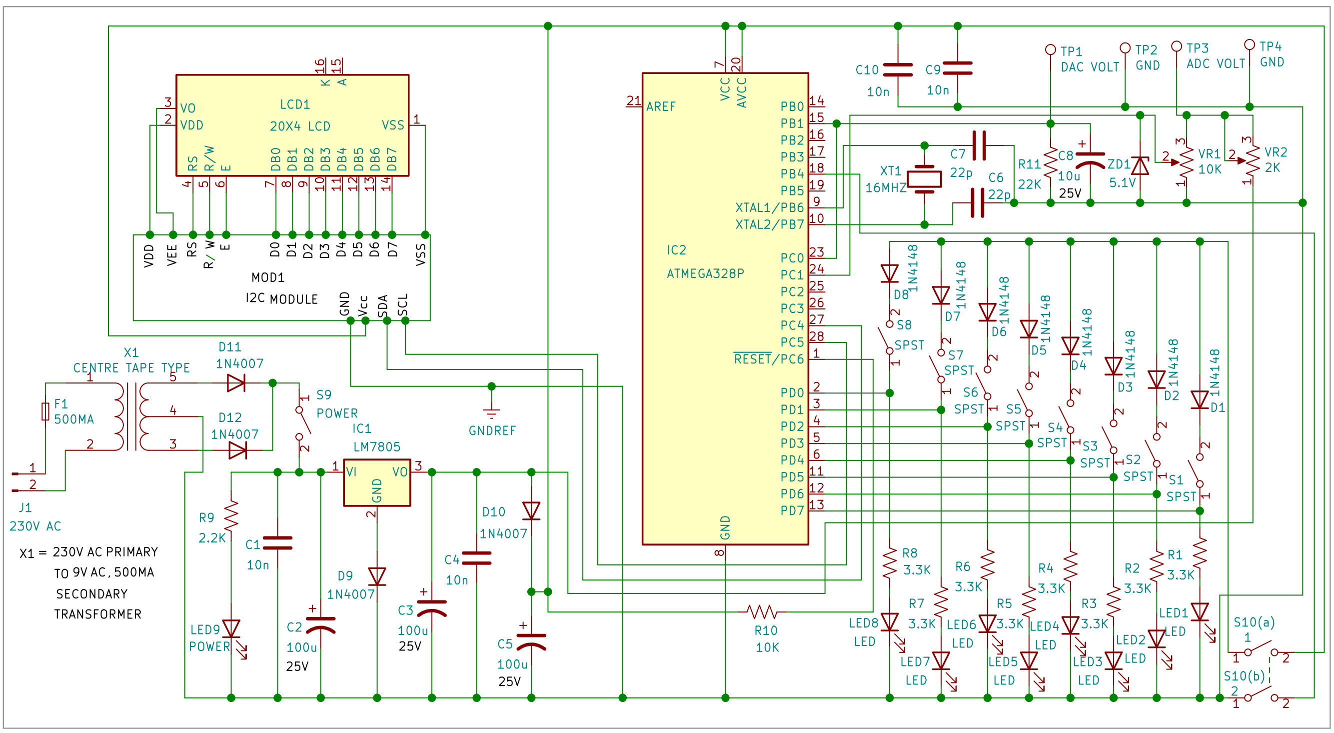

Fig. 1: Circuit diagram of the trainer